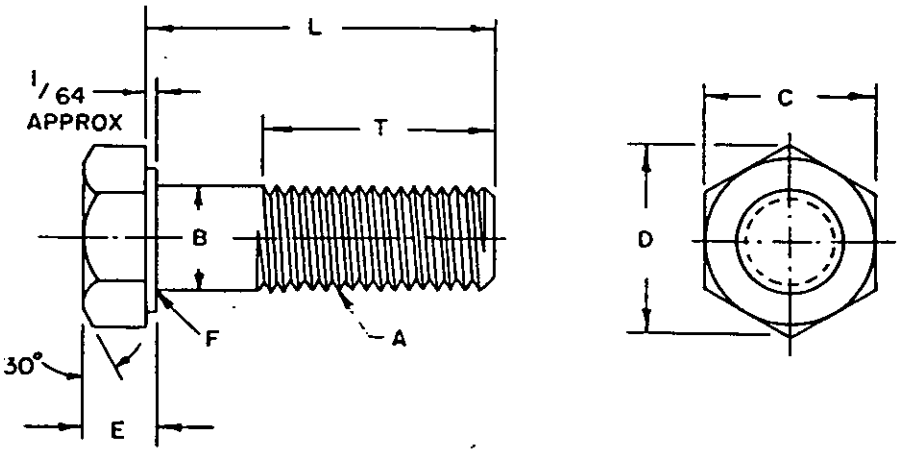
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FED. SUP. CLASS
5306



APPROVED 8 MAY 1957 REVISED Ⓐ 19 AUGUST 1966

(Project 5306-N010Sh) Ⓐ For changes see sheets 1 through 5

CUSTODIANS	OTHER INT.	MILITARY STANDARD	MS 16205 (SHIPS)
Ships	A - N - AF -	BOLTS, MACHINE, REGULAR SEMI-FINISHED HEXAGON HEAD, COPPER-SILICON ALLOY (SILI-	
Procurement Specification Ⓐ MIL-B-857		CON BRONZE), UNC-2A, NON-MAGNETIC	SHEET 1 OF 5

DD FORM 672
1 OCT 52

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PROCUREMENT SPECIFICATION (A) MIL-B-857
DD FORM 672-1 1 OCT 52

CUSTODIANS: Ships
OTHER INT. A - N - AF -

MILITARY STANDARD

BOLTS, MACHINE, REGULAR SEMI-FINISHED HEXAGON HEAD, COPPER-SILICON ALLOY (SILICON BRONZE), UNC-2A, NON-MAGNETIC

SUPERSEDES:

MS16205 (SHIPS)

SHEET 2 OF 5

FED. SUP. CLASS 5306

				1/4		5/16		3/8		7/16	
Nominal size				20-UNC2A		18-UNC2A		16-UNC2A		14-UNC2A	
A Threads per inch				0.250		0.312		0.375		0.437	
B Body diameter		Max.		.217		.276		.334		.394	
		Min.		.437		.500		.562		.625	
C Width across flats		Max.		.425		.484		.544		.603	
		Min.		.505		.577		.650		.722	
D Width across corners		Max.		.484		.552		.620		.687	
		Min.		.163		.211		.243		.291	
E Head height		Max.		.150		.195		.226		.272	
		Min.		.031		.031		.031		.031	
F Radius		Max.		.009		.009		.009		.009	
		Min.									

L Length	Tolerance Size 1/4 - 3/8	Tolerance Size 7/16 - 1/2	Tolerance Size 5/8 - 1-1/4	MS part No.	FIIN	MS part No.	FIIN	MS part No.	FIIN	MS part No.	FIIN
1/4	±1/32	±1/16	±1/8	--		--		--		--	
3/8				1		--		--		--	
1/2				2		25		46		76	
5/8				3		--		47		--	
3/4				4		26		48		77	
7/8				5		--		49		--	
1				6		27		50		78	
1-1/4				7		28		51		79	
1-1/2				8		29		52		80	
1-3/4				9		30		53		81	
2				10		31		54		82	
2-1/4				11		32		55		83	
2-1/2				12		33		56		84	
2-3/4				13		34		57		85	
3				14		35		58		86	
3-1/4				15		36		59		87	
3-1/2				16		37		60		88	
3-3/4				17		38		61		89	
4				18		39		62		90	
4-1/4				--		--		63		--	
4-1/2				--		--		64		--	
4-3/4				--		--		65		--	
5				--		--		66		--	
5-1/2				--		--		67		--	
6				--		--		68		--	
6-1/2	±1/16	±3/32	±3/16	--		--		69		--	
7				--		--		70		--	
7-1/2				--		--		71		--	
8				--		--		72		--	
9				--		--		73		--	
10				--		--		--		--	
11				--		--		--		--	
12				--		--		--		--	
13				--		--		--		--	
14				--		--		--		--	
15				--		--		--		--	
16				--		--		--		--	
18				--		--		--		--	
20				--		--		--		--	

APPROVED 8 MAY 1957 REVISED (A) For changes see sheets 1 through 5

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MIL-B-857

CUSTODIANS: Ships

OTHER INT. A- N- AF-

MILITARY STANDARD

BOLTS, MACHINE, REGULAR SEMI-FINISHED HEXAGON HEAD

COPPER-SILICON ALLOY (SILICON BRONZE), UNC-2A, NON-MAGNETIC

SUPERSEDES:

MS16205 (SHIPS)

SHEET 5 OF 5

Nominal size		1/2	5/8	3/4	7/8
A Threads per inch		13-UNC2A	11-UNC2A	10-UNC2A	9-UNC2A
B Body diameter	Max.	0.500	0.625	0.750	0.875
	Min.	.450	.566	.685	.803
C Width across flats	Max.	.750	.937	1.125	1.312
	Min.	.725	.906	1.088	1.269
D Width across corners	Max.	.866	1.083	1.299	1.516
	Min.	.826	1.033	1.240	1.447
E Head height	Max.	.323	0.403	0.483	0.563
	Min.	.302	.379	.455	.531
F Radius	Max.	.031	.062	.062	.062
	Min.	.009	.021	.021	.021

L Length	Tolerance: Size 1/4 - 3/8	Tolerance: Size 7/16 - 1/2	Tolerance: Size - 5/8 - 1-1/4	MS part No.	FIIN	MS part No.	FIIN	MS part No.	FIIN	MS part No.	FIIN
1/4	±1/32	±1/16	±1/8	---		---		---		---	
3/8				---		---		---		---	
1/2				96		---		---		---	
5/8				---		---		---		---	
3/4				97		130		165		---	
7/8				---		---		---		---	
1				98		131		166		---	
1-1/4				99		132		167		---	
1-1/2				100		133		168		201	
1-3/4				101		134		169		203	
2				102		135		170		204	
2-1/4				103		136		171		205	
2-1/2				104		137		172		206	
2-3/4				105		138		173		207	
3				106		139		174		208	
3-1/4				107		140		175		209	
3-1/2				108		141		176		210	
3-3/4				109		142		177		211	
4				110		143		178		212	
4-1/4				111		144		179		213	
4-1/2				112		145		180		214	
4-3/4				113		146		181		215	
5				114		147		182		216	
5-1/2				115		148		183		217	
6				116		149		184		219	
6-1/2	±1/16	±3/32	±3/16	117		150		185		---	
7				118		151		186		220	
7-1/2				119		152		187		---	
8				120		153		188		221	
9				121		---		189		222	
10				122		155		190		223	
11				123		---		191		224	
12				124		157		192		225	
13				125		158		---		---	
14				---		159		194		---	
15				126		---		195		---	
16				---		161		196		---	
18				---		---		197		---	
20				---		163		---		---	

FED. SUP. CLASS 5309

APPROVED 8 MAY 1957 REVISED (A) For changes see sheets 1 through 5

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DD FORM 672-1 1 OCT 52

PROCUREMENT SPECIFICATION (A) MIL-B-857

CUSTODIANS: Ships

OTHER INT. A- N- AF-

MILITARY STANDARD

BOLTS, MACHINE, REGULAR SEMI-FINISHED HEXAGON HEAD, COPPER-SILICON ALLOY (SILICON BRONZE), UNC-2A, NON-MAGNETIC

SUPERSEDES:

MS16205

(SHIPS)

SHEET 4 OF 5

			1	1-1/8	1-1/4	1-3/8	1-1/2
	Nominal size		8-UNC2A	7-UNC2A	7-UNC2A	6-UNC2A	6-UNC2A
A	Threads per inch						
B	Body diameter	Max.	1.000	1.125	1.250	1.375	1.500
		Min.	.919	1.032	1.157	1.267	1.392
C	Width across flats	Max.	1.500	1.687	1.875	2.062	2.250
		Min.	1.453	1.631	1.812	1.994	2.175
D	Width across corners	Max.	1.732	1.949	2.165	2.382	2.598
		Min.	1.653	1.859	2.066	2.273	2.480
E	Head height	Max.	.657	.719	.813	.878	.974
		Min.	.591	.658	.749	.810	.902
F	Radius	Max.	.093	.093	.093	.093	.093
		Min.	.062	.062	.062	.062	.062

L Length	Tolerance: Size 1/4 - 3/8	Tolerance: Size 7/16 - 1/2	Tolerance: Size 5/8 - 1-1/4	MS part No.	FIIN	MS part No.	FIIN	MS part No.	FIIN	MS part No.	FIIN	MS part No.	FIIN
1/4	±1/32	±1/16	±1/8	---		---		---		---		---	
3/8				---		---		---		---		---	
1/2				---		---		---		---		---	
5/8				---		---		---		---		---	
3/4				---		---		---		---		---	
7/8				---		---		---		---		---	
1				---		---		---		---		---	
1-1/4				231		261		291		---		---	
1-1/2				232		262		292		---		---	
1-3/4				233		263		293		---		---	
2				234		264		294		---		---	
2-1/4				235		265		295		---		---	
2-1/2				236		266		296		---		---	
2-3/4				237		267		297		---		---	
3				238		268		298		---		---	
3-1/4				239		269		299		---		---	
3-1/2				240		270		300		---		---	
3-3/4				241		271		301		---		---	
4				242		272		302		---		---	
4-1/4				243		273		303		---		---	
4-1/2				244		274		304		---		---	
4-3/4				245		275		305		---		---	
5				246		276		306		---		---	
5-1/2				247		277		307		---		---	
6				---		278		308		---		---	
6-1/2	±1/16	±3/32	±3/16	249		---		---		---		---	
7				---		---		---		---		---	
7-1/2				251		---		---		---		---	
8				---		---		---		---		---	
9				253		---		---		---		---	
10				---		---		---		---		---	
11				255		---		---		---		---	
12				---		---		---		---		---	
13				256		---		---		---		---	
14				---		---		---		---		---	
15				---		---		---		---		---	
16				---		---		---		---		---	
18				---		---		---		---		---	
20				---		---		---		---		---	

FED. STOCK CLASS 5305

APPROVED 8 MAY 1957 REVISED (A) For changes see sheets 1 through 5

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FED. SUP. CLASS 5306

(A) Material: Copper silicon alloy (silicon bronze) shall be in accordance with copper alloy 651 of QQ-C-591.

Protective coating: None.

Threads: UNC-2A shall be in accordance with Handbook H-28.

Thread length: Minimum thread length shall be twice the nominal size (diameter) plus 1/4 inch for bolt lengths 6 inches and less, and twice the nominal size plus 1/2 inch for bolt lengths over 6 inches. Bolts too short for minimum thread length shall be thread as close to the head as practicable.

Minimum breaking strength: The minimum breaking strength, in tension, of the bolts shall be as follows:

Sizes to 1/2 inclusive - 80,000 p. s. i.
Sizes 5/8 to 7/8 inclusive - 70,000 p. s. i.
Sizes 1 to 1-1/2 inclusive - 50,000 p. s. i.

(A) Minimum elongation - 10 percent in 2 inches.

Dimensions: All dimensions are in inches, unless otherwise specified.

Notes:

1. In case of conflict with any document referenced herein, this standard shall govern.
2. Referenced documents shall be the issue in effect on date of invitation for bids.

APPROVED 8 MAY 1957 REVISED (A) For changes see sheets 1 through 5

CUSTODIANS	OTHER INT.	MILITARY STANDARD	MS 16205 (SHIPS)
Ships	A - N - AF -	BOLTS, MACHINE, REGULAR SEMI-FINISHED HEXAGON HEAD, COPPER-SILICON ALLOY (SILI-	
Procurement Specification (A) MIL-B-857		CON BRONZE), UNC-2A, NON-MAGNETIC	SHEET 5 OF 5

DD FORM 672
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